

Herb–Drug Interaction Between the Extract of *Hibiscus sabdariffa* L. and Hydrochlorothiazide in Experimental Animals

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ABSTRACT Decoctions of *Hibiscus sabdariffa* L. (Family Malvaceae) are very popular for the preparation of homemade refreshing drinks and are also used medicinally for a variety of ailments. Particularly remarkable are the various scientific reports supporting diuretic and antihypertensive potentials. It is therefore not unusual for patients who are on orthodox antihypertensive medications to use medicinal *H. sabdariffa* drinks concomitantly without regard to the possibility of herb–drug interactions. This possibility necessitated this study in which the pharmacokinetic and pharmacodynamic interactions of *H. sabdariffa* extract (HSE) and hydrochlorothiazide (HCT), a commonly prescribed diuretic drug, were examined. The effects of concomitant administration of HSE on urine volume, urine pH, and urinary concentrations of sodium, bicarbonate, and chloride ions, as well as on the pharmacokinetic parameters of HCT, were determined in experimental rats and rabbits. Co-administration of HSE with HCT caused a significant increase in the volume of urine excreted and resulted in a decrease in the pH of urine and the concentrations of sodium, bicarbonate, and chloride ions. Co-administration of HSE (20–40 mg/kg) with HCT (10 mg/kg) increased and prolonged the plasma concentration, the mean area under the concentration–time curve, and the volume of distribution of HCT achieved over the 24-hour sampling period. The plasma clearance and the elimination rate constant of HCT decreased with increasing dose of HSE co-administered with the HCT. The results of this study reveal a possible herb–drug interaction involving HCT and HSE, used as an ingredient in medicinal or refreshing drinks in many countries.

KEY WORDS: • herb–drug interactions • herbal beverages • *Hibiscus sabdariffa* • hydrochlorothiazide

INTRODUCTION

DRUG INTERACTION REFERS TO the possibility that a drug or nondrug substance may alter the pharmacological effects of another drug given together. Multiple-drug therapy is often necessary and sometimes inevitable in the management of certain disease conditions in order to achieve the desired therapeutic response. In such cases drug interaction is bound to occur and could affect or modify the responses observed. In addition, foods, medicinal foods, and herbs that are taken concomitantly with drugs could also alter the responses to the drug.

The management of hypertension is one of the many disease states where multiple-drug therapy is inevitable and often used to the benefit of the patients. This raises the possibility of drug interactions in the patient. Apart from orthodox medications, some herbal remedies are believed to possess good antihypertensive activities, and the use of these

herbal remedies is on the increase among patients of different educational and socioeconomic status in both developed and low-income countries. Regrettably these herbal therapies are, most often, taken concomitantly with orthodox antihypertensive medications without regard to the possibility of useful or deleterious herb–drug interactions.

Extracts of *Hibiscus sabdariffa* L. (Family Malvaceae) are very popular for the preparation of a homemade refreshing beverage and are also used medicinally for a variety of ailments. *H. sabdariffa* is native to Central and West Africa but grows throughout many tropical areas. This annual herb grows to 1.5 m or higher and produces elegant red flowers. The flowers (calyx and bract portions) are collected when slightly immature. *Hibiscus* has a long history of use in Africa and other tropical countries for many conditions, including hypertension, liver diseases, cancer, constipation, and fever.^{1–4} The fleshy red calyx is used in the preparation of jams, jellies, drinks, and cold and warm teas. The plant is also widely used in Egypt, Iran, and Thailand, as well as in Western countries, and is often found as a component of herbal tea mixtures.⁵

The use of beverages and teas containing *H. sabdariffa* extracts (HSEs) among hypertensive patients is growing

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because of some scientific reports showing that the extract of the plant possesses useful diuretic and hypotensive properties that could be beneficial to them.^{6–10} For example, a trial showed that a 12-day daily treatment of hypertensive patients with a hot aqueous HSE resulted in an average blood pressure reduction of 11/10 mm Hg compared with the untreated control.³ In another comparative study, HSE was shown to possess comparable antihypertensive efficacy to captopril.⁷ This information has fueled an increased level of usage by patients, even among patients who are on orthodox medications, without regard to the possibility of pharmacokinetic or pharmacodynamic herb–drug interactions.

These facts guided our recent studies on the possible herb–drug interactions of HSEs and antihypertensive drugs. A diuretic effect has been identified as one of the possible mechanisms by which *H. sabdariffa* could be exerting its hypotensive activities.³ This study was designed to determine the possibility of herb–drug interactions with the concomitant use of *H. sabdariffa*-containing beverages and hydrochlorothiazide (HCT), a commonly prescribed diuretic. The pharmacokinetic and the pharmacodynamic alterations of the effects of HCT by the HSE were determined in experimental animal models.

MATERIALS AND METHODS

Plant material

Dried samples of the floral parts of *H. sabdariffa* were purchased from a commercial stock in Kaduna Main Market in Northern Nigeria. The purchased sample was properly authenticated by a plant taxonomist, Mr. A. Ozioko of the Bioresources Development and Conservative Program, Nsukka, Nigeria. The dried calyces were pulverized, and a 1-kg portion of the powder was exhaustively extracted by maceration in methanol with intermittent agitation for 96 hours. The extracted solution was evaporated *in vacuo* at reduced pressure to afford a solid HSE. This HSE was subjected to preliminary phytochemical analyses using previously described methods.¹¹

Animals

Adult albino mice (weighing 15–22 g), albino rats (weighing 150–260 g), and local strains of healthy adult rabbits (weighing 1.2–2.0 kg) were used in the study. The animals were of both sexes and were obtained from the Animal House of the Department of Pharmacology & Toxicology, University of Nigeria, Nsukka. The animals were housed under standard conditions ($25 \pm 2^\circ\text{C}$ and 12-hour light/dark cycle). The rabbits were fed with a local grass, *Panicum maxima* Jacq. (Poaceae). The rats and mice were maintained on standard livestock pellets (Livestock feed®, Vital, Lagos, Nigeria) and allowed unrestricted access to drinking water. The use and care of all laboratory animals were conducted in accordance with the internationally acceptable best practices as contained in the European Community Guidelines (EEC Directive of 1986; 86/609/EEC).

Drugs

The HCT tablets (Esidrex®, Novartis, Rueil-Malmaison, France) used were purchased from a community pharmacy. A suspension of HCT (25 mg/mL) was freshly prepared in 10% Tween 80 and used for the study.

Acute toxicity test of the extract

The acute toxicity (50% lethal dose [LD_{50}]) profile of the HSE was estimated in mice using the method described by Lorke.¹² In brief, the tests involved two phases. The first phase is the determination of the toxic range. The mice were divided into three groups of three mice each. The groups were administered the extracts at doses of 10, 100, and 1,000 mg/kg (i.p.), respectively. The treated mice were observed for 24 hours for the number of deaths.

The death pattern in the first phase determined the doses used for the second phase according to the estimation technique of Lorke.¹² In this phase, four mice were given 1,600, 2,900, 3,600, and 5,000 mg/kg, respectively, to each mouse, and the animals were observed for lethality or signs of acute intoxication for 24 hours. The LD_{50} was calculated as the geometric mean of the highest non-lethal dose and the least toxic dose.¹²

Effect of co-administration of HSE and HCT on diuresis

The rats were placed into five groups ($n = 5$) and put singly in single-animal metabolic cages. The animals were denied access to drinking water from 5 hours before the commencement of the experiment until 24 hours after the experiment. Group I was given only distilled water (q.s.). Group II was given HCT (10 mg/kg) alone, Group III received HCT (10 mg/kg) + distilled water (q.s.), Group IV received HSE (40 mg/kg) alone, and Group V received HCT (10 mg/kg) + HSE (40 mg/kg). In addition to these treatments, each rat was also given a single load of normal saline (5 mL/100 g, p.o.) using an intragastric tube. The volume of urine excreted by each animal at 1, 2, 4, 8, and 24 hours following the various treatments was collected in a test tube and measured using a measuring cylinder. At the end of the 24-hour observation, the urine samples collected from each animal at the various time points were pooled together, and the pH, specific gravity, and Na^+ , Cl^- , and HCO_3^- concentrations were determined.

Effect of HSE on pharmacokinetics of HCT

Nine rabbits were randomized into three groups ($n = 3$): Group I was given HCT (10 mg/kg) alone, Group II was given HCT (10 mg/kg) + 20 mg/kg HSE, and Group III was given HCT (10 mg/kg) + 40 mg/kg HSE. All drug treatments were by the oral route. At intervals of 0.5, 1, 2, 4, 8, and 24 hours blood samples were drawn from the marginal ear vein of the rabbit. The blood samples were deproteinized by the addition of two drops of acetonitrile immediately after collection. The blood samples were centrifuged at 1,000 g for 15 minutes, after which the absorbance values were determined at 323 nm. A drug-free plasma sample was used as the blank solution.

TABLE 1. EFFECT OF *H. SABDARIFFA* ON THE DIURETIC EFFECTS OF HYDROCHLOROTHIAZIDE

Treatment group	Urine volume (mL)				
	1 hour	2 hours	4 hours	8 hours	24 hours
Control (distilled water)	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.77 ± 0.26	4.03 ± 0.96
HCT (10 mg/kg)	0.90 ± 0.30	1.70 ± 0.29 ^a	2.77 ± 0.62 ^{ac}	3.17 ± 0.57	4.57 ± 0.67
Distilled water + HCT (10 mg/kg)	0.40 ± 0.21	0.90 ± 0.21	1.97 ± 0.27	3.68 ± 0.92	6.73 ± 1.03
HSE (40 mg/kg)	0.00 ± 0.00	0.00 ± 0.00 ^b	0.60 ± 0.00	2.23 ± 1.03	3.73 ± 1.33
HSE (40 mg/kg) + HCT (10 mg/kg)	2.15 ± 0.72 ^{abc}	3.57 ± 0.64 ^{abc}	5.337 ± 1.54 ^{abc}	7.77 ± 1.79 ^{abc}	9.60 ± 1.99 ^{ab}

Data are mean ± SEM values.

^aSignificantly different relative to distilled water alone control treatment.

^bSignificantly different relative to hydrochlorothiazide (HCT) (10 mg/kg) treatment.

^cSignificantly different relative to HCT (10 mg/kg + distilled water, q.s.) treatment.

HSE, *H. sabdariffa* extract.

Beer–Lambert's plot

A 5 mg% solution of standard HCT (Novartis) was prepared in 0.1 N NaOH and scanned between a range of 200 and 400 nm to determine the wavelength of maximum absorption for the drug, which was 323 nm. A standard calibration curve was plotted for HCT between 1 and 8 mg% at 323 nm. This curve was used in determining the concentration of HCT in plasma samples.

Calculation of pharmacokinetic parameters

We assumed a one-compartment model after a single oral administration of the drug. The following pharmacokinetic properties were determined for each animal, and the mean values were calculated.

Area under the curve. The area under the curve (AUC) measures the amount of drug that reaches the systemic circulation after a specific dose and is directly proportional to the amount of drug absorbed. The average plasma concentration of HCT was plotted against respective time, and the AUC was calculated using the trapezoid rule.¹³

Elimination rate constant. The elimination rate constant (K_{el}) was obtained from the slope of the least square regression analysis for the apparently linear portion of the log serum concentration versus time curve using the following relationship: Slope = $-0.43 \times K_{el}$.

Half-life. Assuming a first-order elimination rate, the half-life ($T_{1/2}$) of the drug in the serum obtained from each phase of the study was determined using the following relation:¹⁴ $T_{1/2} = 0.693 \times K_{el}^{-1}$.

Peak serum concentration and time to peak serum concentration. Peak serum concentration and time to peak serum concentration were obtained from the serum concentration versus time curve.

Serum clearance rate. The serum clearance rate (CL) of HCT in all the phase of the study was determined using the following relation:¹⁴ $CL = K_{el} \times V_d$, where V_d is the volume of distribution.

V_d . This was calculated using the following relationship:¹⁴

$$V_d = \frac{D_b}{K_{el} \times AUC}$$

where D_b is the amount of drug administered.

RESULTS AND DISCUSSION

Co-administration of HSE with HCT caused a significant increase in the volume of urine excreted, as well as a significant decrease in pH and urine Na^+ , HCO_3^- , and Cl^- concentrations (Tables 1 and 2). The increase in urine vol-

TABLE 2. EFFECT OF *H. SABDARIFFA* EXTRACT AND HYDROCHLOROTHIAZIDE ON THE PHYSICOCHEMICAL PARAMETERS OF URINE SAMPLES

Treatment group	pH	Specific gravity	Electrolyte concentration (mmol/L)			
			[Na ⁺]	[K ⁺]	[Cl ⁻]	[HCO ₃ ⁻]
Control (distilled water)	7.63 ± 0.12	1.04 ± 0.01	431.34 ± 67.71	338.84 ± 52.08	53.33 ± 12.02	160.00 ± 15.28
HCT (10 mg/kg)	9.28 ± 0.09	1.03 ± 0.00	616.21 ± 36.49	375.09 ± 39.38	293.33 ± 29.33	143.33 ± 8.82
Distilled water + HCT (10 mg/kg)	9.26 ± 0.02	0.99 ± 0.00	297.23 ± 36.25	321.10 ± 19.89	198.67 ± 19.33	76.67 ± 6.67
HSE (40 mg/kg)	7.73 ± 0.12 ^b	1.01 ± 0.12 ^a	329.44 ± 20.0 ^b	352.38 ± 12.59	55.00 ± 12.58 ^b	70.67 ± 5.81 ^{abc}
HSE (40 mg/kg) + HCT (10 mg/kg)	8.60 ± 0.10 ^{bc}	1.02 ± 0.00 ^c	366.10 ± 29.67 ^b	300.49 ± 16.09	62.67 ± 15.59 ^b	73.33 ± 8.82

Data are mean ± SEM values.

^aSignificantly different relative to distilled water alone control treatment.

^bSignificantly different relative to HCT (10 mg/kg) treatment.

^cSignificantly different relative to HCT (10 mg/kg + distilled water, q.s.) treatment.

TABLE 3. EFFECT OF *H. SABDARIFFA* EXTRACT ON THE PHARMACOKINETICS OF HYDROCHLOROTHIAZIDE IN RABBITS

Treatment	AUC ([mg/mL]×hour)	C _{max} (mg/mL)	T _{max} (hours)	T _{1/2} (hours)	K _{el}	CL (mL/kg/hour)	V _d (L/kg)
HCT (10 mg/mL)							
Alone	0.658 ± 0.003	0.140 ± 0.0	4.0 ± 0.0	1.32 ± 0.00	0.524 ± 0.005	5.3 ± 0.05	10.08 ± 0.011
+ HSE (20 mg/kg)	2.178 ± 0.001	0.170 ± 0.0	6.1 ± 0.0	2.19 ± 0.003	0.316 ± 0.006	3.38 ± 0.05	10.67 ± 0.028
+ HSE (40 mg/kg)	2.389 ± 0.008	0.142 ± 0.0	6.1 ± 0.0	8.97 ± 0.01	0.077 ± 0.006	3.02 ± 0.01	39.10 ± 0.057

Pharmacokinetic data were calculated from using drug concentration values measured in collected blood samples at different time points by assuming a simple one-compartment model for each animal, and the mean values were recorded. Data are mean ± SEM values.

AUC, area under the curve; CL, serum clearance rate; C_{max}, maximum serum concentration; T_{1/2}, half-life; T_{max}, time to C_{max}; K_{el}, elimination rate constant; V_d, volume of distribution.

ume was recorded from the 1st hour up until the 24th hour. The combination of HSE and HCT also caused a decrease in urinary Na⁺ concentration compared with the group administered HCT alone. The extract did not cause an attendant increase in the urinary excretion of Na⁺, Cl⁻, and HCO₃⁻. These effects suggest a possible benefit of HSE in cases of dilutional hyponatremia. The tendency of the extract to cause a decreased excretion of Cl⁻ and a decrease in pH suggests the possibility of metabolic acidosis with prolonged usage.

Consistent with an earlier report on the safety of *H. sabdariffa*-containing beverages,¹⁵ the LD₅₀ of the extract was estimated to be greater than 5 g/kg in our study. This may imply that the HSE is reasonably safe without any evidence of acute toxicities¹² when taken alone as a beverage. Co-administration of HSE (20–40 mg/kg) with HCT (10 mg/kg) increased and prolonged the plasma concentration of HCT achieved over the 24-hour sampling period. As shown in Table 3, most of the pharmacokinetic parameters of HCT evaluated were increased by the addition of HSE in the regimen. The mean concentration–time AUC is 0.658 when HCT was administered alone, but increased significantly in all the other treatment groups where HCT was administered in combination with HSE. The maximum AUC was recorded when 40 mg/kg HSE was administered with the HCT. Because the AUC indicates the total amount of drug achieved in the body, the increased AUC caused by HSE co-administration suggests a possible retention of HCT.

CL, which is the volume of blood cleared of drug per unit time,¹⁴ decreased with increasing dose of HSE co-administered with the HCT. The reductions in the CL means that less of the HCT will be cleared from the body at any given time with increasing dose of HSE given together. The V_d depicts the pattern of distribution of drugs between intracellular and extracellular compartments.¹⁴ In our study, the V_d of HCT increased with increasing dose of HSE administered together. The K_{el} of HCT decreased with increasing dose of HSE. Because K_{el} reflects the rate at which drug is removed from the body, it implies that the HSE slows the rate HCT elimination from the body.

The study established significant alteration in both the pharmacodynamics and the pharmacokinetic indices of HCT caused by HSE when administered concurrently. Although there is evidence supporting possible antihyper-

tensive activities of *H. sabdariffa*-containing beverages,^{3,16} patients should avoid concomitant usage of a *H. sabdariffa*-containing herbal beverage with HCT diuretics in the management of hypertension.

AUTHOR DISCLOSURE STATEMENT

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